

Output Content (OC)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: MILU | Context: North Indian Hindi-Medium Postgraduate Teacher — MCQ Evaluation Deployment

Definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Justification

Annotation relied on subject experts at online exam portals [Q26] plus AI4Bharat volunteer audit [Q87], with manual cluster-to-subject assignment [Q41]. The paper does not provide annotator demographic details (regional background, Hindi-medium vs. English-medium, subject expertise) and does not report inter-annotator agreement. Web search confirmed no published source resolves this gap (gap_id 6). The institutional base (IIT Madras + IBM Research India) does not guarantee Hindi-medium North Indian regional representation. The user's confirmation (elicitation Q4) that competitive-exam keys are broadly consistent with North Indian teacher norms substantially mitigates concern, supported by dataset evidence that answer keys for India-centric items appear standard. However, for items requiring regionally specific judgments (e.g., Hindi literary canon, regional history framing) the absence of annotator-demographic transparency leaves a moderate gap.

Strengths

- Subject experts at exam portals manually tagged questions and ensured answer accuracy [Q26].
- AI4Bharat volunteer audit layer adds quality assurance [Q87].
- Manual cluster-to-subject label assignment rather than fully automated [Q41].
- User confirmation that evaluation schemas are broadly consistent across North Indian boards and competitive exams reduces label-disagreement risk for typical items.

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Partially. Ground-truth labels derive from competitive-exam answer keys [Q26], which the user confirmed are broadly compatible with North Indian teacher norms (elicitation Q4). However, no Hindi-medium North Indian stakeholder-perspective validation is documented.

OC-2: Disagreement risk is moderate: typical India-GK items show good alignment, but for culturally specific items (canonical Hindi literature framing, regional history interpretations) the absence of annotator-demographic data prevents firm assessment.

OC-3: INSUFFICIENT DOCUMENTATION — no Datasheet or Data Statement section provides annotator demographics. Institutional affiliation (IIT Madras, IBM Research India) is documented [Q9, Q10] but does not guarantee Hindi-medium North Indian representation. Web search confirmed this gap remains unresolved (gap_id 6).

ID	Evidence
WEB-10	NAACL 2025 paper confirms institutional base (IIT Madras + IBM Research India) but does not publish annotator demographics.

OC-4: Re-annotation by Hindi-medium North Indian teachers would strengthen convergent validity, particularly for Arts & Humanities, Hindi Literature, and regional history items. Not currently performed.

OC-5: INSUFFICIENT DOCUMENTATION — no inter-annotator agreement statistics are reported, so aggregation/erasure of minority perspectives cannot be assessed.

OC-6: Convergent-validity concern is moderate due to undocumented annotator demographics, partially mitigated by user-confirmed schema consistency. External validity for culturally specific items remains uncertain.

Information Gaps

- Annotator demographic profile (regional background, language medium, subject expertise).
- Inter-annotator agreement statistics for Hindi items.
- Whether any Hindi-medium North Indian teachers were involved in audit.

Requires Expert Verification

- Hindi-medium North Indian teacher review of a sample of culturally specific items (literature, regional history, civics) to assess label consistency with regional norms.

Remediation

Gap 1: No published annotator demographics or inter-annotator agreement; representation of Hindi-medium North Indian educators in annotation/audit unknown.

Recommendation: Recruit a small panel of Hindi-medium North Indian board teachers (3-5 per state across UP/MP/Rajasthan/Bihar) to re-annotate a stratified sample of 200-500 culturally specific items (Hindi Literature, regional history, regional civics) and measure label agreement with MILU keys.